

Lean 4 proof trees

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By the Curry–Howard correspondence, a natural-deduction *proof tree* is the same object as a Lean 4 *proof term*: each inference rule is a term constructor, and discharging an assumption is a λ -abstraction.

$\wedge$ -introduction. From a and b , conclude $a \wedge b$.

$$\frac{a \quad b}{a \wedge b} \wedge\text{I}$$

Lean: `<ha, hb>` (i.e. `And.intro ha hb`).

$\rightarrow$ -elimination (modus ponens). From $a \rightarrow b$ and a , conclude b .

$$\frac{a \rightarrow b \quad a}{b} \rightarrow\text{E}$$

Lean: `h ha` (apply the arrow to its argument; the `apply` tactic).

$\rightarrow$ -introduction / discharge. Proof of $a \rightarrow (b \rightarrow a)$; superscripts mark the discharged assumptions.

$$\frac{\frac{[a]^1}{b \rightarrow a} \rightarrow\text{I}^2}{a \rightarrow (b \rightarrow a)} \rightarrow\text{I}^1$$

Lean: `fun (ha : a) (_ : b) => ha`.

A two-branch derivation. $(a \wedge b) \rightarrow (b \wedge a)$.

$$\frac{\frac{[a \wedge b]^1}{b} \wedge\text{E}_2 \quad \frac{[a \wedge b]^1}{a} \wedge\text{E}_1}{\frac{b \wedge a}{(a \wedge b) \rightarrow (b \wedge a)} \wedge\text{I}} \rightarrow\text{I}^1$$

Lean: `fun h => <h.right, h.left>`.